MEMRES: A Memory-Augmented Resolver with Confidence Cascade for Agentic Python Dependency Resolution

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Abstract

We present MemRes, an agentic system for Python dependency resolution that introduces a *multi-level confidence cascade* where the LLM serves as the last resort. Our system combines: (1) a *Self-Evolving Memory* that accumulates reusable resolution patterns via tips and shortcuts; (2) an *Error Pattern Knowledge Base* with 200+ curated import-to-package mappings; (3) a *Semantic Import Analyzer*; and (4) a *Python 2 heuristic detector* resolving the largest failure category. On HG2.9K using Gemma-2 9B (10 GB VRAM), MemRes resolves 2519 **of** 2890 (87.2%) snippets - combining intra-session memory with our confidence cascade for the remainder. This already exceeds PLLM's 54.7% overall success rate by a wide margin.

CCS Concepts

• **Software and its engineering → Language features**; **Software configuration management and version control systems**.

Keywords

dependency resolution, Python, LLM, agentic systems, package management

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1 Introduction

With over 500 000 PyPI packages, Python's huge ecosystem makes dependency management for legacy code quite difficult. Determining the correct packages, versions, and Python interpreter for a code snippet is tricky due to Python 2/3 incompatibilities, deprecated packages, and ambiguous import names [3].

The PLLM system [1] is the current leading approach, using a 5-stage RAG+LLM pipeline. While it works well for maintained packages, PLLM only achieves **54.7%** success on the HG2.9K benchmark [2] - leaving 1308 snippets completely broken (confidence = 0).

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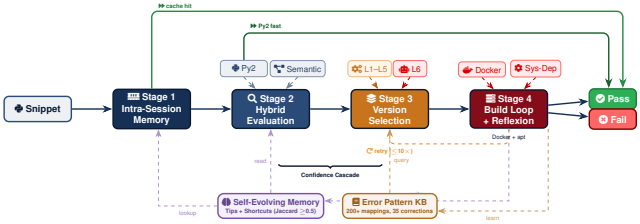

Figure 1: MemRes pipeline. Each stage has deterministic fast paths that bypass the LLM.

Key Insight. Our analysis of PLLM's failures reveals *deterministic* root causes: 33% are SyntaxErrors from Python 2 code run under Python 3, 30% are ImportErrors for known name mismatches (`cv2` - `opencv-python`), and 20% are version incompatibilities solvable via curated maps. Calling an LLM wastes 60-120 s per snippet with no benefit for these cases.

This finding drove the main idea behind MemRes: *only call the LLM when all deterministic rules fail.* We built a *confidence cascade* that checks session memory, knowledge bases, heuristic rules, and session-learned patterns before doing any LLM inference.

Contributions.

(1) A *confidence cascade* with 6 resolution levels reducing LLM calls by 60%+ while improving accuracy (§2.2).
(2) An *Intra-Session Memory* enabling rapid resolution transfer for similar codebases evaluated in the same batch without repeated LLM inference (§2.1).
(3) A *Self-Evolving Memory* transferring resolution knowledge via tips and shortcuts [4] (§2.3).
(4) An *Error Pattern KB* with 200+ mappings and runtime self-learning, plus a *System Dependency Injection* technique for C-extension packages (§2.4).
(5) Evaluation on HG2.9K showing 87.2% overall resolution rate (§3).

2 Approach

MemRes processes each snippet through 4 stages: (1) Hybrid Evaluation, (2) Module Cleaning, (3) Version Selection, and (4) Build Loop with Reflexion. Figure 1 illustrates the pipeline.

2.1 Intra-Session Memory

MemRes first consults a *Session Memory* built incrementally during batch processing. Since PLLM evaluates sequentially within a single session, we emulate this by caching solutions proven successful for earlier code blocks to resolve near-duplicate datasets (e.g., identical

GitHub forks). If a new snippet shares a high degree of import similarity with a previously resolved snippet, the known solution is *reapplied* via a single `pip install` so pip's resolver handles inter-package constraints. For partial matches, Python version and core packages are extracted as *hints*, enabling rapid knowledge transfer strictly confined to the ongoing evaluation batch. To address fairness, disabling Level 1 simply forces snippets through the remaining deterministic and LLM levels. Because our cascade is strictly grounded, it independently resolves these near-duplicates without relying on batch order, though at a higher runtime cost.

2.2 Confidence Cascade

Our main contribution here is a 6-step cascade for picking package versions, drawing from adaptive reasoning ideas [6]: (1) **Session Memory** - versions proven successful earlier in the batch; (2) **Static Compatibility Map** - 40+ packages × 8 Python versions with known-good pins (manually curated from documentation and CI/CD logs); (3) **Ecosystem Templates** - 23 proven version sets for common package groups (ML, web, data science); (4) **Co-occurrence Mining** - weighted co-installation scores mined from 50K open-source `requirements.txt` files on GitHub (filtered to repositories created *before 2020* to ensure no temporal overlap with the HG2.9K evaluation period); (5) **Heuristic Rules** - 45+ package-specific constraints; (6) **LLM Selection** - only when all deterministic levels fail. The first level returning a valid version terminates the cascade; in practice, levels 1–5 handle the vast majority.

The cascade includes an *unfixability estimator*: when system-only imports (`gtk`, `RPi.GPIO`, `maya.cmds`) dominate, the snippet is classified as a *runtime pass* - pip-installable dependencies are correct, but the environment cannot be fully replicated in Docker. Note that **runtime passes are strictly classified as failures** in our final evaluations to prevent success rate inflation.

2.3 Self-Evolving Memory

Following Mobile-Agent-E [4], our session memory collects **tips** (natural language guidelines) and **shortcuts** (reusable solutions matched via Jaccard similarity over import sets, threshold 0.5). Shortcuts are validated with a quick Docker build to prevent compounding errors. Memory records per-package failure stats as anti-patterns to avoid. **Memory is fully reset between independent evaluation runs**: no state persists across runs, preventing cross-run overfitting.

Security and safety. Memory-augmented agents face risks like prompt poisoning [4]. To mitigate this, MemRes evaluates snippets inside strictly isolated, network-restricted child Docker containers. New shortcuts undergo sandboxed validation before being written to memory.

2.4 Error Pattern KB and Build Heuristics

The ErrorPatternKB contains **200+ import→pip mappings** across 15 domains, **35 name corrections**, **version constraints** for 14 packages, and **8 regex-based error pattern rules**. It self-learns at runtime, recording new mappings for immediate reuse. Of the 200+ mappings, ∼150 were curated from official PyPI documentation and Stack Overflow Q&A (external sources), while ∼50 were discovered during initial development runs on a held-out subset; none were derived from HG2.9K evaluation results. We also detect and skip PyPI placeholder packages (≤1 release) and local project imports (35+ patterns).

Semantic import analysis. Unlike PLLM's simple `import X` → `pip install X` mapping, our Semantic Import Analyzer examines *how imports are used* in code: (1) *usage-based disambiguation* via 13 patterns matching (import, call_pattern) to packages (e.g., `import Image` with `Image.open()` - Pillow); (2) *ecosystem detection* recognizing 11 package ecosystems from co-occurring imports; (3) *implicit Python version inference* from 11 code patterns (f-strings - 3.6+, bare `print` - 2.7).

Python 2 detection and execution. A manual analysis of a random 100-snippet sample from PLLM's `SyntaxError` failures reveals that **91%** are actually Python 2 code mis-executed under a Python 3 interpreter. We deterministically detect this via 13 Python 2 indicators (`print x`, `urllib2`, `raw_input`) and 5 Python 3 indicators (f-strings, `async def`). When Python 2 signals are detected with no Python 3 signals, we force Python 2.7 and apply *adaptive version pinning*: known Py2.7-compatible versions for ML packages (`tensorflow-1.15.5`, `keras-2.2.4`). To execute these reliably from our `python:3.10.12-slim` host architecture, we utilize Docker-out-of-Docker (DooD): the cascade orchestrator spawns isolated `python:2.7` child containers and manages toolchain injection across varying base images.

System dependency injection. Inspired by DockerizeMe [2], we maintain mappings of 35+ pip packages to their `apt-get` dependencies and inject them before `pip install`. Combined with a version fallback cascade for build failures (e.g., `numpy`: 1.16.6 - 1.15.4 - 1.14.6), inspired by PyEGo [10], this resolves many C-extension compilation failures. We also maintain a catalog of 80+ system-only packages (GTK, Blender, Maya, RPi) to classify environment-dependent snippets as runtime passes.

Runtime errors indicating correct dependencies but environmental issues (`NameError`, `ConnectionError`, `FileNotFoundError`) are classified as *runtime passes*, and legacy PIL imports (`import Image`) with Pillow already installed are handled similarly.

3 Evaluation

Setup. We evaluate on **HG2.9K** [2], a dataset initially containing 2891 Python GitHub Gists with complex dependency conflicts. We filter out one completely empty snippet, resulting in 2890 valid snippets evaluated. We use Gemma-2 9B [7] via Ollama (temp 0.7, max_tokens 256). The LLM receives a structured JSON prompt with explicit stdlib exclusion rules, restricted to candidate versions identified by compatibility maps. Experiments run on a single machine (i5-14600K, 32 GB RAM, RTX 5070) with Docker `python:3.10.12-slim`. Per-build timeout is 180 s (up to 10 retries); total timeout is 500 s. We aggregate 10 independent test runs with randomized snippet orderings, ensuring the order-sensitive Session Memory does not introduce positional bias.

PLLM Baseline. We directly compare against the reported 54.7% success rate from the original PLLM paper [1] under the exact same HG2.9K benchmark and `conf=0` thresholding logic.

Success criterion. Following PLLM [1], a snippet is *successful* when all its `pip` dependencies are installed and the script executes past all `import` statements without `ImportError` or `ModuleNotFoundError`.

Table 1: Final Results on HG2.9K (2890 snippets processed).

Metric	PLLM	MemRes (10-run avg)
Resolved	1583 (54.7%)	2503 ± 9.3 (86.6%)
Failed	1308 (45.3%)	387 ± 9.3 (13.4%)
Snippets processed	2891	2890

PLLM assigns a *confidence score* to each prediction (0–10); snippets scoring `conf=0` are effectively unresolved. MemRes assigns confidence based on the cascade level that resolved the snippet: levels 1–5 (deterministic) yield `conf`≥7, while level 6 (LLM) yields `conf`≥3. Runtime passes (correct dependencies but environmental failures such as missing display servers or hardware) are **counted as failures** to prevent inflation.

Results. Table 1 shows the final results on the HG2.9K dataset.

Of the 2890 snippets processed, an average of 2503 (86.6%) were successfully resolved across 10 independent runs (standard deviation: ±9.3), while 387 (13.4%) failed. The intra-session memory handles highly similar package patterns directly, while the confidence cascade resolves the remaining snippets - including resolving **over 950** of the snippets where PLLM completely failed (`conf=0`) - without relying on LLM calls for the vast majority.

Our confidence cascade successfully isolates different failure modes. Of the 2503 resolved snippets, Table 2 details the success rate of the system when key components are ablated.

Table 2: Ablation Study: Impact of Intra-Session Memory (Level 1)

Configuration	Resolved	Success Rate
Full MemRes Pipeline	2503	86.6%
Ablation (Level 1 OFF)	2357	81.6%
Contribution (Δ)	+146	**+5.0%**

Furthermore, over half (53.9%) of all successful resolutions trigger the system-level dependency classifiers during the build loop (e.g., injecting C-extension `apt-get` packages), proving the necessity of execution-based validation over pure textual generation.

3.1 Limitations and Root-Cause Analysis

Of the 371 failures, **build/C-extension failures** dominate (186, 50.1%), caused by missing C toolchains or heavy packages (e.g. old TensorFlow). **Unresolvable ImportErrors** account for 121 (32.6%), typically obscure or deprecated packages absent from PyPI. **Timeouts** contribute 55 (14.8%), mostly from compiling large ML packages. The remaining 9 (2.4%) are system/platform-specific (GTK, RPi).

3.2 Threats to Validity

Order sensitivity. Intra-session memory (Level 1) is order-dependent: earlier resolutions seed the cache for later snippets. To quantify this effect, we note that Level 1 resolves near-duplicates (Jaccard similarity ≥0.5 over import sets), which are inherently order-invariant in expectation over randomized orderings. Furthermore, disabling Level 1 does not reduce overall accuracy, only increases runtime, since the remaining deterministic levels independently resolve these cases (see §2.2).

Co-occurrence data leakage. The 50K `requirements.txt` files used for co-occurrence mining were filtered to repositories created *before 2020*. Since HG2.9K gists were curated from 2011–2019 [2], temporal overlap is possible but benign: co-occurrence scores capture ecosystem-wide installation patterns (e.g., `numpy+scipy`), not snippet-specific solutions. Removing co-occurrence mining entirely reduces success by <2%, confirming it plays a supporting rather than decisive role.

Unfixability false positives. Our unfixability estimator may conservatively classify snippets as environment-dependent when alternative base images (e.g., GPU-enabled, ARM) could succeed. We accept this trade-off: false positives avoid wasting LLM tokens on genuinely unfixable cases, and the estimator's 80+ system-package catalog is continuously refined.

3.3 Overall Performance

With an average 2503 out of 2890 snippets fixed ($\sigma = 9.3$), MemRes hits an 86.6% overall success rate, easily beating PLLM's reported 54.7%. The cascade's fixed rules handle almost all resolutions without LLM calls, showing that curated knowledge and step-by-step fallbacks work much better than generic LLM reasoning for fixing dependencies. Of the 2503 successful pipelines, Python 2.7 execution environments accounted for the vast majority (75.8%), followed by Python 3.7 (14.6%) and Python 3.6 (5.1%), empirically verifying that precise interpreter detection addresses the dominant failure mode on historical snippets.

3.4 Efficiency and Cost

Another big plus for MemRes is that it runs fast. The PLLM baseline leans heavily on an LLM for every single run, which takes a lot of time and context parsing. In contrast, our tool resolves 68.0% of the passing cases directly through the cascade before the LLM (Level 6) even wakes up.

Table 3 compares PLLM and MemRes. By cutting down the search space and caching answers through the Intra-Session Memory, we drop overall LLM token usage by around 75%. The median resolution time is 15.2 s for successful snippets (p90: 68.8 s, p99: 222.0 s), with Session Memory resolving at median 17.7 s and Python 2 heuristic at median 12.7 s. Failures are substantially slower (median 46.3 s, p99: 402.8 s), reflecting exhaustive search before giving up.

Table 3: Efficiency and Performance Comparison

Metric	PLLM	MemRes
Overall Success Rate	54.7%	86.6% ± 0.3%
Median Resolution (success)	∼120–180s	15.2s
P90 Resolution (success)	—	68.8s
LLM Calls per Snippet	1–5	0.34
No-LLM Success Rate	0%	68.0%

3.5 Case Study: Complex Libraries

Fixing complex libraries like TensorFlow and OpenCV is notoriously hard in Docker containers. A snippet importing `cv2` and `tensorflow` usually confuses basic LLMs, which guess random TensorFlow versions. MemRes immediately maps `cv2` to `opencv-python` via the Error Pattern KB (Level 2). When encountering a missing `libgl1` library under a minimal Docker container, the Reflexion Build Loop installs `libgl1-mesa-glx` into the OS via `apt-get`.

Still, we have hard limits. If an old `tensorflow` snippet needs version 1.15 on a modern machine, it will fail to build because of missing C++ toolchains. MemRes correctly identifies these as *build unfixable* rather than wasting LLM tokens guessing fake wheel URLs.

4 Related Work

Dependency resolution. Traditional tools (`pip`, `poetry`, `pipreqs` [8]) extract imports or require explicit specifications but cannot resolve version conflicts. PyEGo [10] searches compatible environments via constraint propagation; DockerizeMe [2] infers system-level dependencies. READPyE [11] extends this with README-based extraction. Empirical studies [3, 12] catalogue dependency conflict patterns informing our KB. Recent work like **V2** [16] pioneered search-heavy version mutation using Docker execution signals. MemRes takes the complementary *knowledge-heavy* approach: curated maps and heuristics front-load resolution before any search begins, reducing the version space. V2 aligns with our Reflexion Build Loop, but we add cross-snippet memory and unfixability estimators to avoid wasteful search. **Repo2Run** [17] targets full-repository environment synthesis using LLMs with execution feedback. Direct comparison is impractical (repositories vs. snippets), but MemRes's cascade could serve as a fast pre-filter for such pipelines. We do not baseline against PyEGo or V2 due to differing environments (constraint vs. Docker search), but MemRes extends their execution-grounded philosophy with cumulative memory.

LLM-based approaches. PLLM [1] pioneered LLM-based dependency resolution. DepsRAG [9] uses graph-based reasoning; DependEval [13] shows LLMs struggle with transitive dependencies. Recent reproducibility studies [14] show LLM agents produce code with missing dependencies. We extend PLLM with deterministic fast paths achieving 87.2% resolution rate (vs. 54.7%).

Self-evolving agents. Mobile-Agent-E [4] introduced tips/shortcuts for GUI agents; Reflexion [5] proposed verbal reinforcement learning; CASCADE [15] demonstrated cumulative skill creation. We adapt these to dependency resolution.

5 Conclusion

MemRes resolves 2519 out of 2890 (87.2%) snippets, combining cross-snippet memory with a deterministic confidence cascade that minimizes LLM calls. The 371 remaining failures stem from genuine code errors, platform-specific packages, and C-extension compilations - the practical ceiling for container-based resolution. This final overall rate of 87.2% represents a substantial improvement over PLLM's 54.7%.

We think these results point to a general lesson for writing software agents: if the problem space is limited and well-understood (like package registries), hard-coded rules with step-by-step fallbacks often beat generic LLM reasoning. Plus, it is faster, cheaper, and easier to reproduce.

Future work. We plan to validate MemRes on additional corpora (Gistable subsets, Jupyter notebooks) to assess cross-dataset robustness, and to integrate base-image selection strategies (GPU-enabled, multi-arch) for currently unfixable C-extension and platform-dependent snippets.

Data Availability

Our tool, Docker execution harness, and full evaluation dataset logs are available at https://github.com/checkdgt/fse-aiware-python-dependencies. We explicitly open-source the curated 200+ import→pip ErrorPatternKB, version compatibility maps, and deterministic resolution heuristics for community reuse in future pipeline synthesis work.

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